Matteo Falasca

MSc student in theoretical physics, University of Trieste
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EDUCATION

Dec 2026 (expected) — **MSc in Physics**, University of Trieste. Taught in English.
Weighted average 28.4/30; all exams completed on schedule.

Dec 2024 — **BSc in Physics**, University of Trieste.

RESEARCH

Master's thesis — **Graviton Scattering and Induced Decoherence**

Supervisor: Prof. Angelo Bassi

- Effective field theory within canonical QFT, aimed at many-body open-system decoherence.
- Perturbative spin-2 gauge field coupled to matter with an original algebraic analysis.
- Derived graviton–matter cross sections that generalize the light–matter case and include velocity corrections, which I could not find in the literature.
- Ongoing: gravity-induced decoherence from scattering and comparison with condensed matter system decoherence from novel analysis.

Bachelor's thesis — **Calibration of Galaxy Cluster Miscentering Distribution from eROSITA Data**

Supervisor: Prof. Alexandro Saro — code

- Built a neural-network emulator that predicts functions with no analytic form from a large simulated dataset, used it to model position errors.
- Parameter space too large to analyze directly, implementation of ingenious techniques such as Latin hypercube sampling.

PROJECTS

Numerical project — **Locality transfer matrix for noisy quantum circuits** — code

- Built a faster algorithm for the matrix of Grossi et al. (arXiv:2410.01893), which describes how a noisy evolution mixes different subspaces of information.
- Such algorithm could also parametrize where information flows in a generic many-body system.
- Written in Python within a collaboration and later used in their research.

Course project — **The minority game** (6 ECTS)

- Simulated a model of many interacting agents and compared it with its analytical reduction to a mean-field oscillator affected by white noise.

Numerical project — **Epidemic growth** — code

- Extrapolated Italian COVID-19 data.
- Confronted with epidemic models, then analyzed and usefully represented them.

SELECTED COURSES grades out of 30

Light–matter	QFT I (28), QFT II (27), Atoms, Molecules and Photons (28), thesis on spin-2–matter interactions (40 ECTS)
Condensed matter	Quantum Many-Body Systems (28), Statistical Mechanics (27), Quantum Statistical Mechanics (26).
Simulation and data	Computational Physics Lab (30), Machine Learning (29), Data Modelling (28, BSc).

SCHOOLS

18 Aug – 5 Sep 2025	**ICTP School on Quantum Dynamics of Matter, Light and Information**, Trieste. Three-week selective school on quantum many-body physics, quantum optics and light–matter interactions. Lectures by Subir Sachdev on quantum critical dynamics and the Sachdev–Ye–Kitaev model and its simulations.
11 – 16 May 2025	**Bad Honnef Physics School "Frontiers of Quantum Metrology for New Physics Searches"**, DPG and WE-Heraeus Foundation. Quantum metrology and light–matter interactions. Lectures by Silke Ospelkaus on the control of dipolar molecules.

SKILLS

Programming	Fortran (advanced), Python, bash, LaTeX.
Languages	Italian (native), English (fluent).

TEACHING AND OTHER ACTIVITIES

Oct 2025 – Mar 2026	**Tutor for Bachelor students**, University of Trieste. University-funded individual and group sessions for first-year students on how to study for physics exams.
Since 2021	Private lessons in physics and mathematics, also for university exams.
Since 2017	**Football referee** (FIGC), now in Eccellenza, the top regional division. Taught me advanced teamwork and effective communication under pressure
2017 – 2019	President of my Rotary Junior Club. Helped me build advanced social skills.
2020	High-school merit scholarship. 400€

REFEREES

Prof. Angelo Bassi	Thesis supervisor, University of Trieste. Contact on request.
Prof. Francesco Scazza	University of Trieste. Contact on request.